
AN00021-EC NB-IoT AT Command Development User Guide_V1.1

EigenCOMM Wireless Microcontroller

Document Description

The purpose of present document is to help customers to implement their own AT commands.

Contents

1.	Introduction	3
1.1	Purpose	3
2.	Overview	3
2.1	AT server Architecture.....	3
2.2	AT CMD Execute Flow	4
2.3	File Architecture	5
3.	Add New AT CMD	7
3.1	Step to add new AT CMD.....	7
3.2	Example to Add Synchronous Command.....	7
3.3	Example to Add Asynchronous command.....	9
4.	Notes.....	10
5.	Version.....	10
6.	About US.....	11

1. Introduction

1.1 Purpose

This document describes the AT command implementation supported by Eigencomm AT server module. The purpose of this document is to help customers to implement their own AT commands.

2. Overview

2.1 AT server Architecture

The architecture of AT command server is shown in Figure 1. It has 6 components.

AT command received from serial interface is parsed by AT CMD Parser, then parser will find matching handler function by searching a table which lists all the defined at commands.

The AT command handler function which is called execute function will execute the command. The execute function could handle and response directly or send a signal to a service task. The service task will process the received signal and sends confirm(response) signal back to AT task, and then AT task calls response function to process the received signal and forward the response to serial interface. Service task could also send unsolicited signal to AT task, AT task will call indication function and forward the unsolicited message to serial interface.

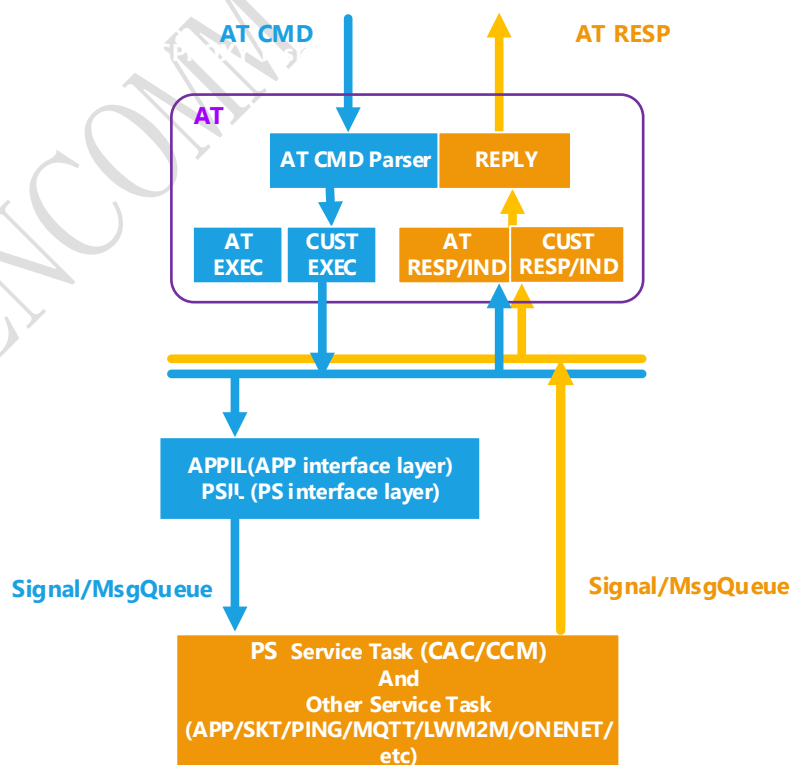


Figure 1 AT server architecture

Components 1 AT CMD Parser:

After receiving AT command from serial interface, AT CMD parser will parse the received command, mainly some syntax parsing work according to specification of AT command. If there is syntax error, it will return ERROR directly; otherwise, it will call the specific AT command execution function to process the AT command. The specific AT command execution function is found from a AT commands table. Figure 2 shows the contents of AT CMD table.

CMD Name	Parameter attribute	Execute function	Response function	Indication function
----------	---------------------	------------------	-------------------	---------------------

Figure 2 AT command table contents

Components 2 EXEC MODULE:

This module will extract and check AT command's parameters, and do specific processing according to the feature of the command.

Separate to 3 parts: AT PS , AT CUST and AT REF.

AT PS executes PS related AT commands, mainly 3GPP AT commands.

AT CUST executes APPL AT commands, such as MQTT, LWM2M and etc. This part is open source in SDK.

AT REF executes some special commands, such as Implement the same commands as other NB-IoT modules.

All the AT functions are listed in three AT commands tables.

Components 3: APPIL and PSIL MODULE

This module is the interface of APP layer and PS layer.

It implemented some APPL/PS APIs, these APIs are called by EXEC functions and send signal to APP/PS service tasks if necessary.

Components 4: Service Task

This module maintains service tasks, which will process specific signal sent form AT task, handle it and sends response or indication signal to AT task.

Components 5: RESP/IND MODULE

This module processes confirm and indication signal from service tasks. It implements response function and indication function for specific AT command, the functions are listed in at command table.

Components 6: REPLY MODULE

This module forward the string format response or indication (unsolicited message) to serial interface.

2.2 AT CMD Execute Flow

AT command execution flow is shown in the figure 3.

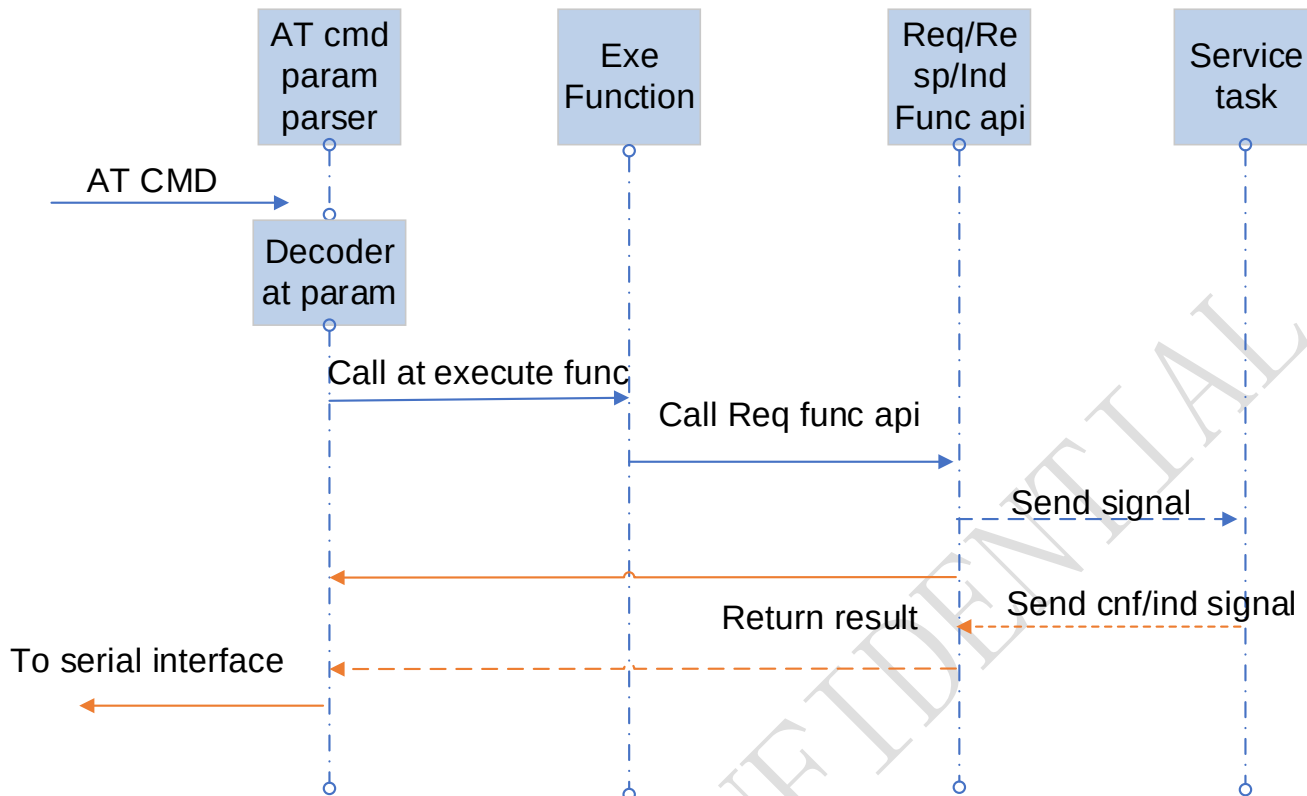


Figure 3 at command execute flow

2.3 File Architecture

Figure 4 and 5 show the file architecture of AT server.

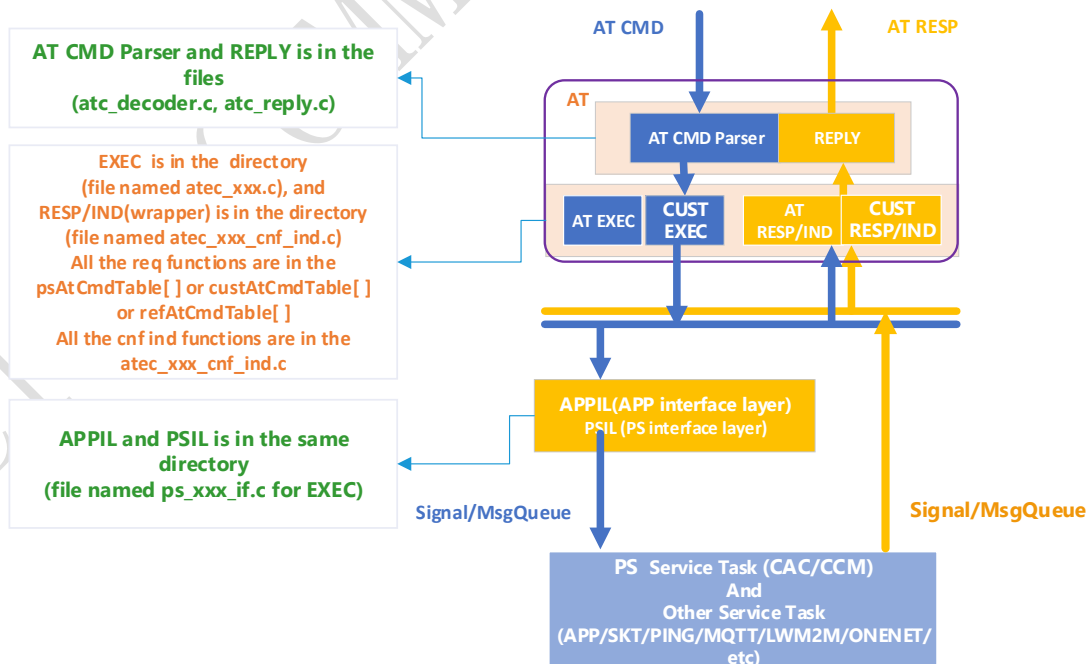


Figure 4 file architecture

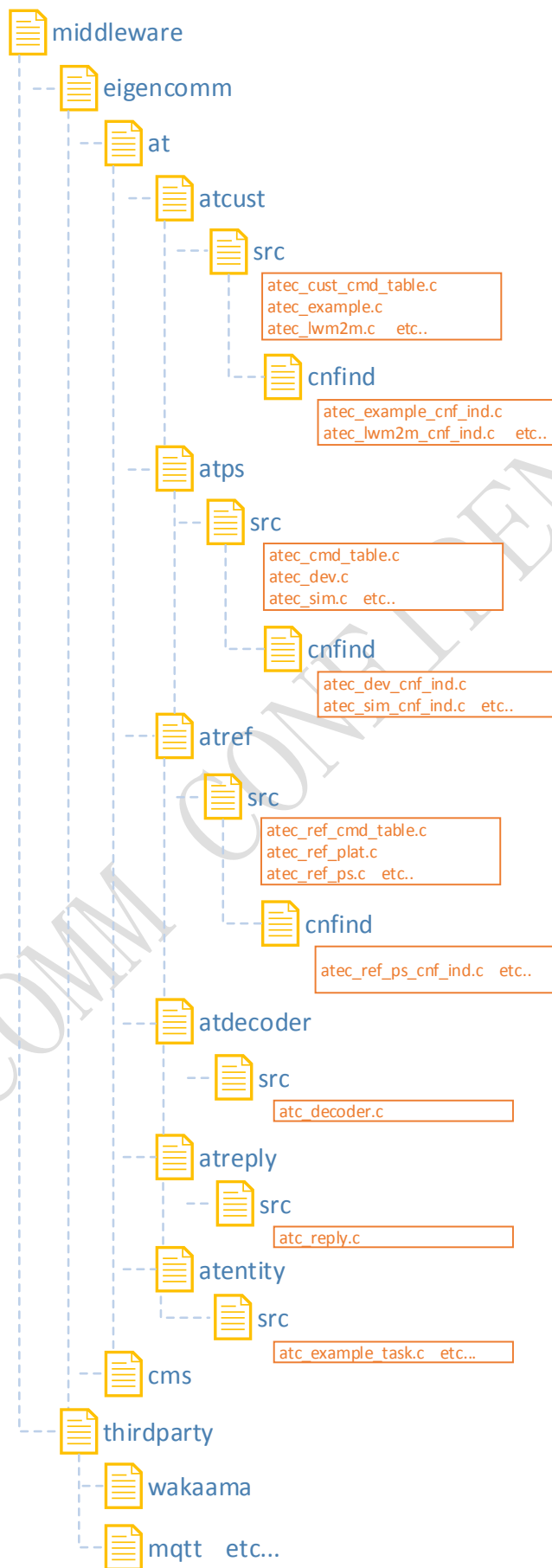


Figure 5 files tree

To add a new AT command, related handle function including execute function, response function, and indication function, should be added to at commands tables, such as psAtCmdTable[], custAtCmdTable[] and refAtCmdTable[], and all the functions should be implemented in the atec_XXX.c. API function should be added if necessary. For execute API function, it should be added in xx_XXX_if.c, such as ps_XXX_if.c.

3. Add New AT CMD

3.1 Step to add new AT CMD

Step 1: confirm the feature of AT command.

Step 2: confirm name of AT command.

Step 3: determine the number of parameters and parameter attributes (parameter type, optional / mandatory). Which will be fulfilled into parameter attribute structure as part of AT command table.

Step 4: confirm timeout of AT command and fulfilled into the at commands table which will be used by AT parser.

Step 5: confirm AT command is synchronous or asynchronous. The decision may be affected by: the complication of the new AT command, whether it needs further processing by service task or simply handled directly.

If synchronize command, it is handled directly in the execute function, no need to send signal to service task.

So, only need to implement execute function.

If asynchronous command, it need to send a signal to a service task which will further handle the command.

So need to implement both execute and response functions.

All work to do is: fill the AT command table with command name, parameter attributes, timeout value and execute/response function. The detail feature of execute/response function is the main effort of the implementer.

3.2 Example to Add Synchronous Command

The steps below show how to add a synchronize AT command.

Eigencomm NB-IoT SDK provide code example to add a synchronous AT command.

The “AT+TESTA” is a synchronous command. It is implemented in atec_example.c.

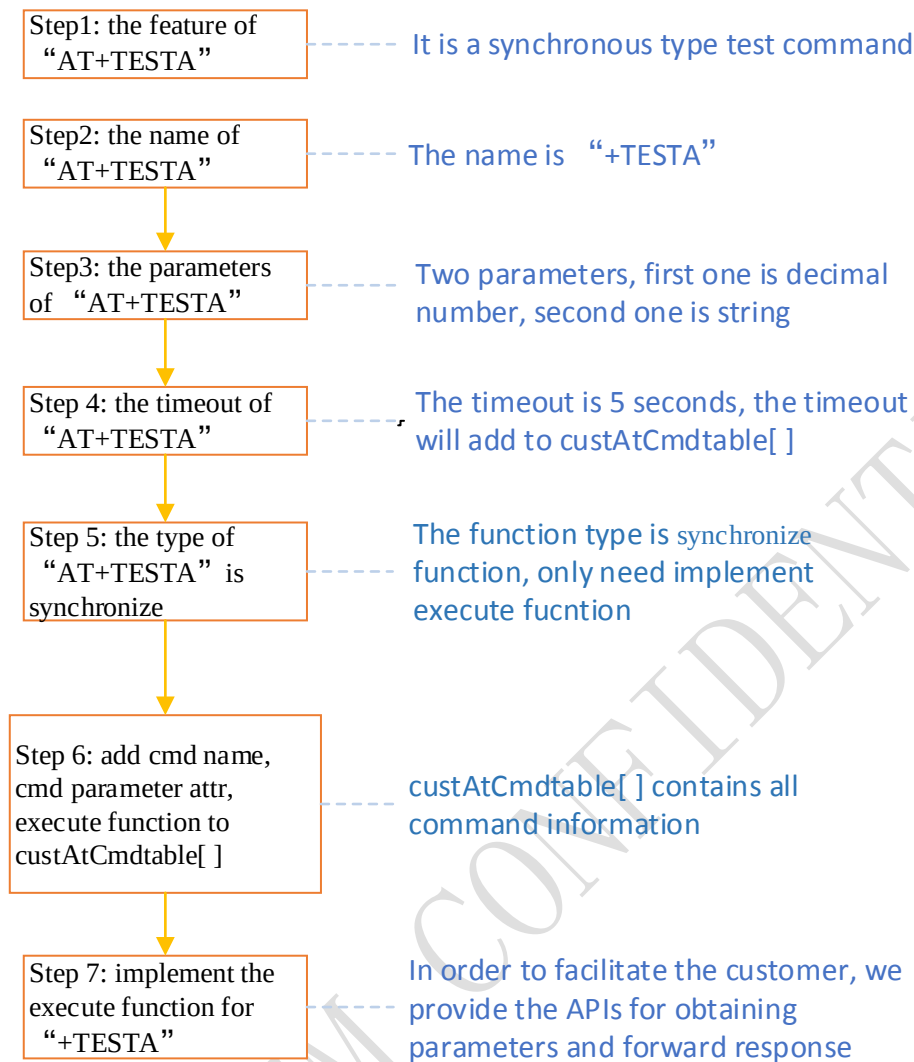


Figure 6 add synchronous command flow

The test example "+TESTA" is shown below, the command responses "+TESTA: REQ".

```

AT+TESTA=1,test
+TESTA: REQ
OK
  
```

Figure 7 synchronous command test result

3.3 Example to Add Asynchronous command

For asynchronous command, it will send signal to service task, after handle the signal, service task will send back a signal to AT task as a response.

Eigencomm NB-IoT SDK provide code example to add an asynchronous AT command.

The “AT+TESTB” is an asynchronous command. It is implemented in atec_example.c.

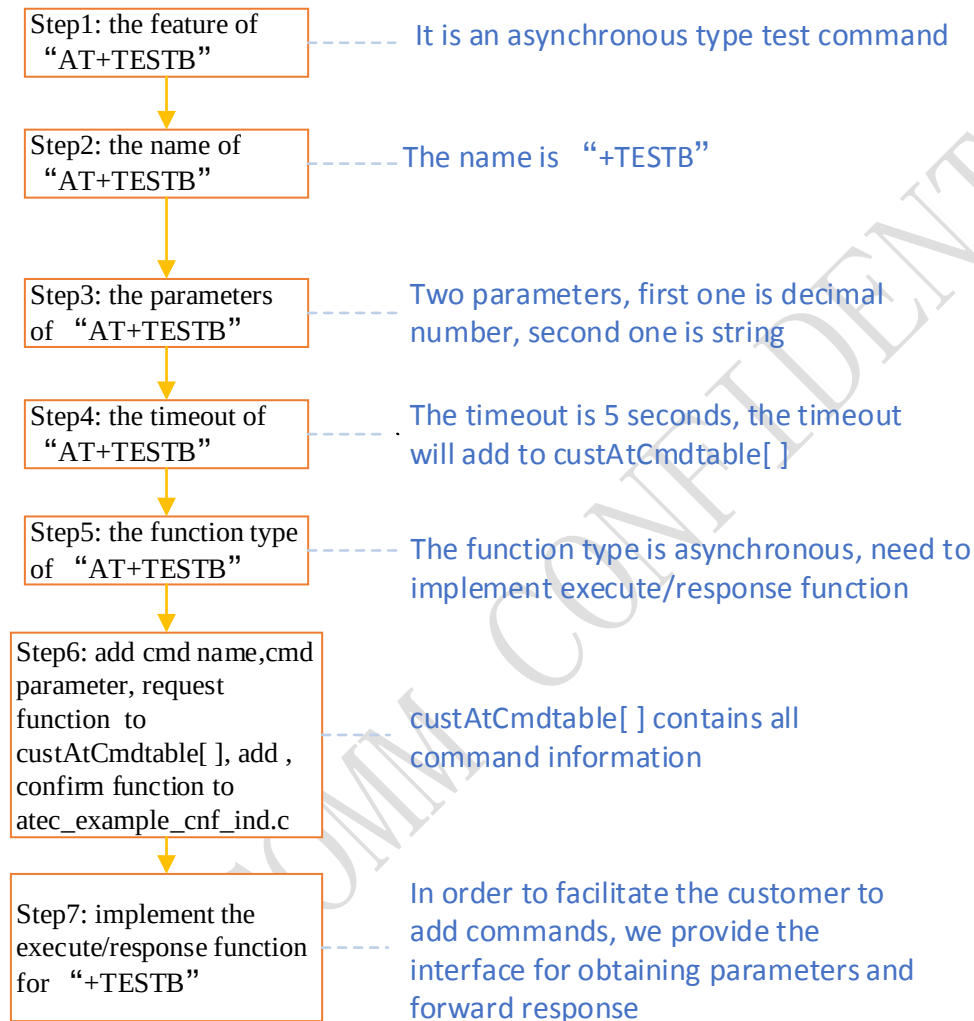


Figure 8 add asynchronous command flow

The test example “+TESTB” is as follow

```

AT+TESTB=1,test
+TESTB: CNF
OK
  
```

Figure 9 asynchronous command test result

4. Notes

(1) when the execute function is implemented, the parameter is got one by one, so the parameter index cannot be mistaken

5. Version

Version	Date	Comments
Draft	2019-11-11	
V1.0	2019-11-28	
V1.1	2020-02-20	

6 About US

Shanghai EigenCOMM Technology Co.,Ltd is established in Feb 2017 in Zhangjiang Hi-Tech park, Shanghai China (www.eigencomm.com) . EigenCOMM focuses on cellular based IoT chipset and solution. With accumulation of experience over the years in algorithm, L1/L2/L3 protocol, SoC, RF , platform & application software as well as the communication architecture and system, team makes its way to IoT industry.

The ultra-low power NB-IoT chipset EC616(S) has already enter mass production stage. The 4G Cat.1 chipset EC618 is in engineering samples stage, 5G is also in the R&D stage.

Eigencomm Technoligies Ltd.

Address: Room 707, No 298 Xiangke Road, Shanghai, China

Zip Code: 201210

Phone:

E-mail:

Website: <http://www.eigencomm.com>

Support Window

E-Mail: support@eigencomm.com